

COURSE TITLE : WORKING DRAWING I
COURSE CODE : 4187
COURSE CATEGORY : A
PERIODS/WEEK : 3
PERIODS/SEMESTER : 39
CREDITS : 2

TIME SCHEDULE

Module	Topics	Period
1	Foundation and masonry.	12
2	joinery	9
3	Vertical transportation	9
4	Roof trusses	9
TOTAL		39

Course Objective :

Sl.	Sub	Student will be able to
1	1	Prepare working drawings of foundations , masonry , joinery details, vertical transportation and steel trusses.
	2	Understand the specification of building materials
	3	Prepare detailed drawings of specified parts of the above topics.

After completing this course student should be able to:

Module I : Foundation and masonry

1.1.0. Draw detailed sections of different types of footings for walls.

- 1.1.1. Draw plan and sections of R.C.C column.
- 1.1.2. Draw Plan and section of raft foundation
- 1.1.3. Draw Details of pre cast and cast in situ piles
- 1.1.4. Draw Pile cap details of heavy concrete and steel columns

1.2.0. Draw Plans of odd and even courses of English and Flemish bonds of one and one and half thick brick walls

1.3.0. Draw Plans and elevation of rat trap bond

1.4.0. Draw Plans of odd and even courses of corners and T joints of one brick thick brick wall

Module II: Joinery

2.1.0. Identify the components of door and window

2.1.1. Draw details of wooden panelled door

2.1.2. Draw details of half glazed door with wooden frame

2.1.3. Draw details of glazed windows in wood and metal.

Module III: Vertical transportation

3.1.0. Set up stair lay outs of different types to suit different situations

3.1.1. Prepare working drawings of RCC stairs showing reinforcement details

3.2.0. Prepare detailed plans and sections of a passenger lift

Module IV : Trusses

4.1.0. Prepare working drawings of steel trusses with tubular and angular sections

4.1.1. Draw corrugated sheet fixing details

CONTENT DETAIL

MODULE-I- FOUNDATION AND MASONRY

R.C.C. Column-raft foundation-pre-cast pile-cast-in –situ pile-pile cap details of heavy concrete and steel columns- Plans of odd and even courses of English and Flemish bonds of one and one and half thick brick walls-rat trap bond - Plans of odd and even courses of corners and T joints of one brick thick wall -

MODULE-II—JOINERY

Panelled door-half glazed door with wooden frame-glazed windows in wood and metal

MODULE-III – VERTICAL TRANSPORTATION

R.C.C Stair with reinforcement details-passenger lift

MODULE-IV—TRUSSES

Steel trusses with angular and tubular sections—details of corrugated sheet fixing

REFERENCE

1. B.T.S.Prabhu, A.Achuthan - Building Drawing and Detailing
2. S.P.Arora,S. P. Bindra - The text book of building construction
3. W.B.Mckay - Building construction